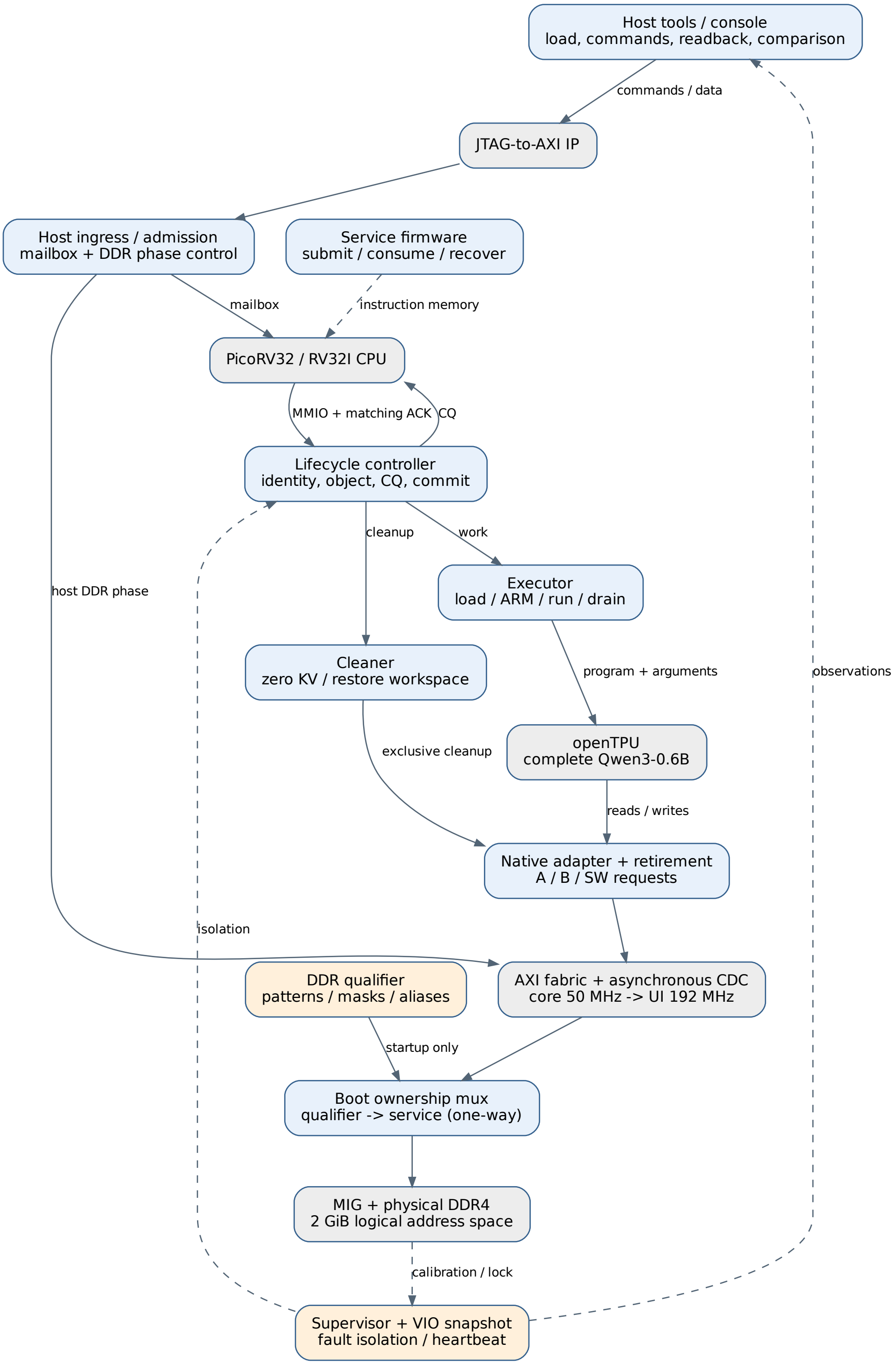RISK-KV: shared-prefix service architecture